

driving cars must be able to operate safely in diverse conditions, including heavy traffic and adverse weather. Furthermore, they should adapt to unexpected situations, such as road closures or accidents.

Ethical implications are also a concern in the realm of self-driving cars. For example, these vehicles may face decisions that involve balancing the safety of passengers with the safety of other road users like pedestrians or cyclists. This raises complex ethical questions that must be carefully considered and addressed.

Several companies are currently working on developing self-driving car technology.

Waymo (owned by Alphabet Inc., Google's parent company), Tesla, Uber, General Motors Cruise (San Francisco), Baidu (China and the United States), Aptiv (Las Vegas) and Mobileye (a subsidiary of Intel and Apple) are some of the key players who have already released their self-driving cars and are evolving the tech.

As the technology continues to advance, we can expect to see more companies entering the market leading to further progress towards fully autonomous vehicles.

The architecture of self-driving cars

The architecture of a self-driving car is complex, consisting of multiple subsystems, sensors, and algorithms.

The human-machine interface acts as the user interface, enabling the car's occupants to interact with the vehicle and control its actions. It incorporates features like touch screens, voice commands, and physical controls (see Figure 1).

The architecture of a self-driving car is constantly evolving, and new technologies and approaches are being developed to improve its safety, reliability, and efficiency. As these technologies continue to advance, we can expect to see more widespread adoption of self-driving cars in the future.

AI and machine learning

Self-driving cars use a combination of AI techniques, including deep learning, reinforcement learning, machine learning, and neural networks to analyse and interpret data from sensors and make decisions about how to navigate the car safely and efficiently.

Deep learning employs neural networks with multiple layers to

analyse and interpret complex data. This is particularly effective in tasks such as object recognition and image classification, which are crucial for self-driving cars.

Reinforcement learning assists self-driving cars in learning from their environment through trial-and-error interactions. It enables training the car to make decisions based on its current state and desired outcomes, such as safely navigating to a destination.

Machine learning encompasses various techniques for analysing and interpreting data. It plays a vital role in processing data from sensors and making decisions about how to safely and efficiently navigate the car.

Neural networks are used in self-driving cars to analyse and interpret data from sensors and enable decisions about how to navigate the car.

Machine/deep learning algorithms used in self-driving cars

Self-driving cars employ several algorithms, each playing a crucial role in enabling the vehicle to navigate its environment safely and efficiently. Here are some of the key algorithms used by the systems shown in Figure 1.

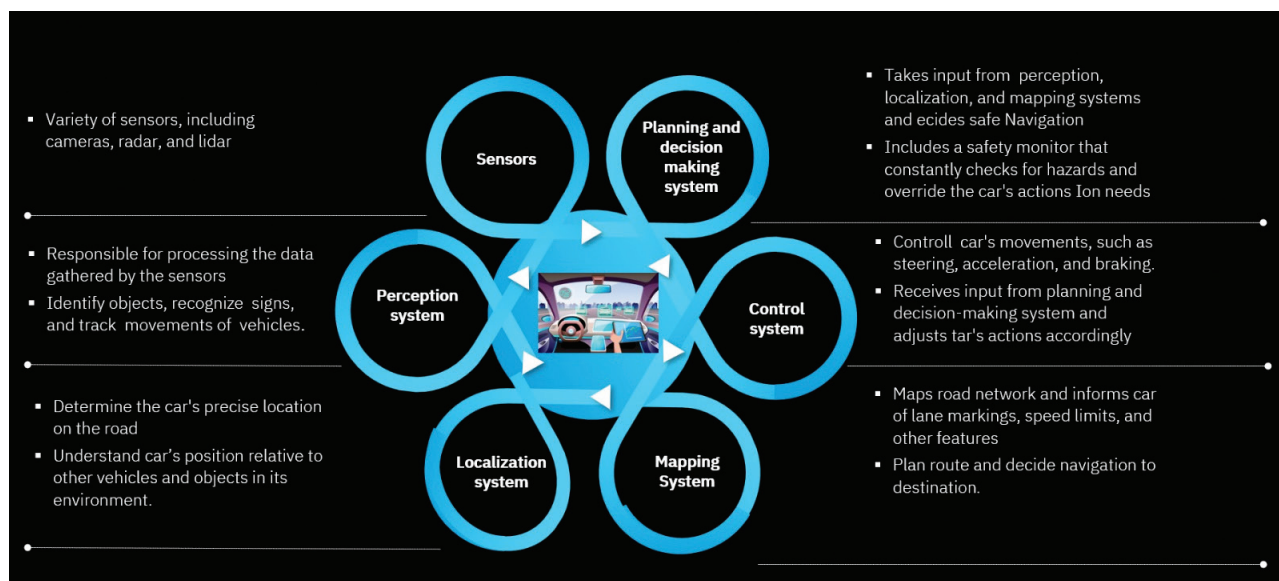


Figure 1: The architecture of a self-driving car (Photo credits: freepik.com)